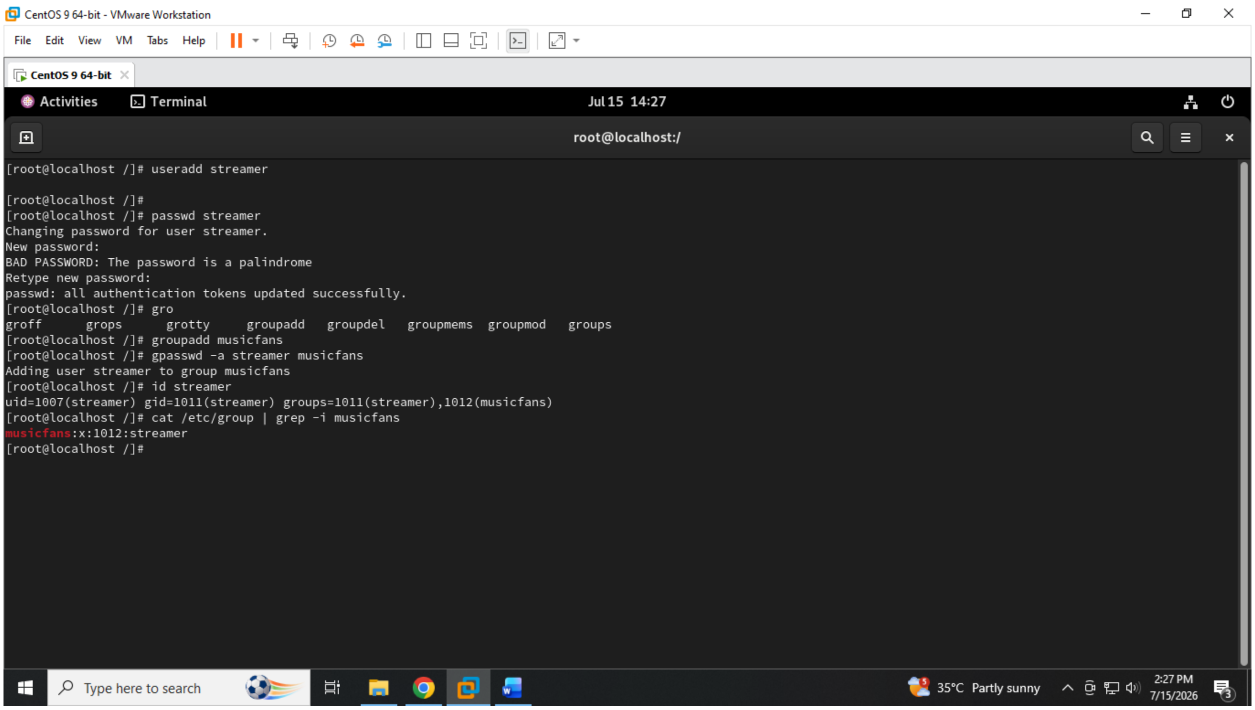
MODULE-7 : MANAGING USERS & SSH SECURITY & UNDERSTANDING OF SELINUX

GUIDED BY : VIPUL MISTRI

Q.1 Create a new user called streamer and a group called musicfans on your Linux system, then add streamer to the musicfans group. Verify the group membership using the id command.

ANS :



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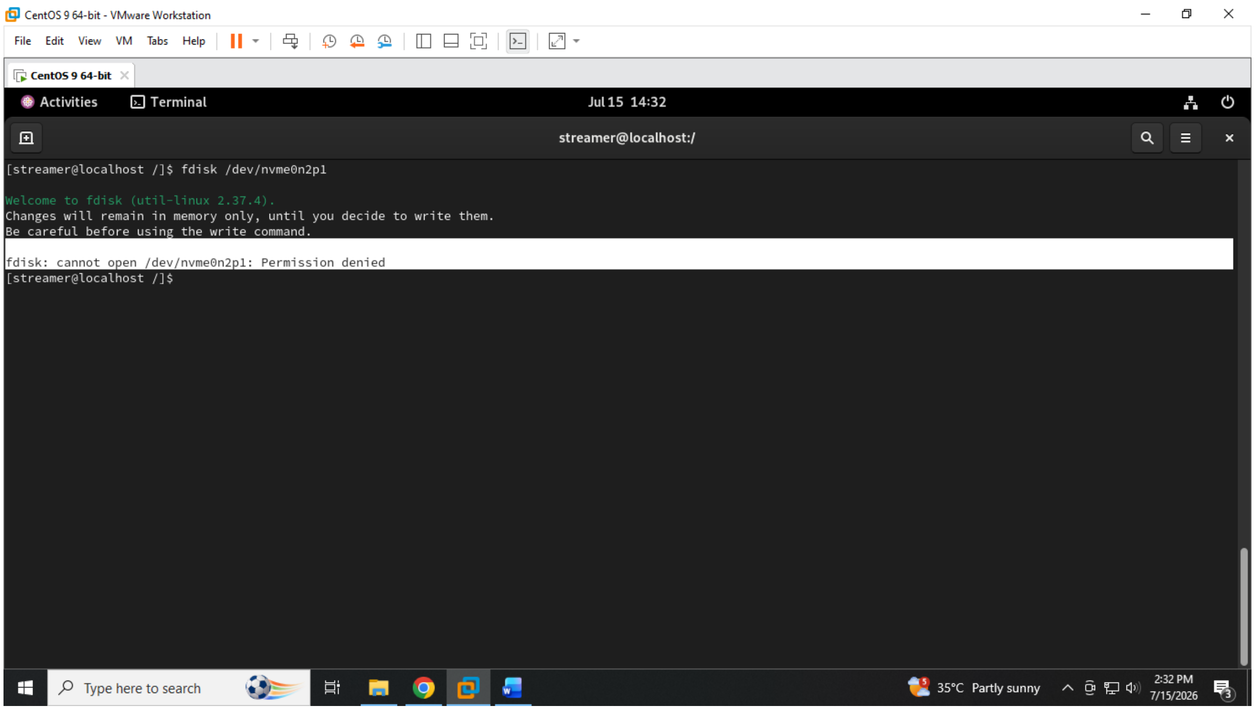
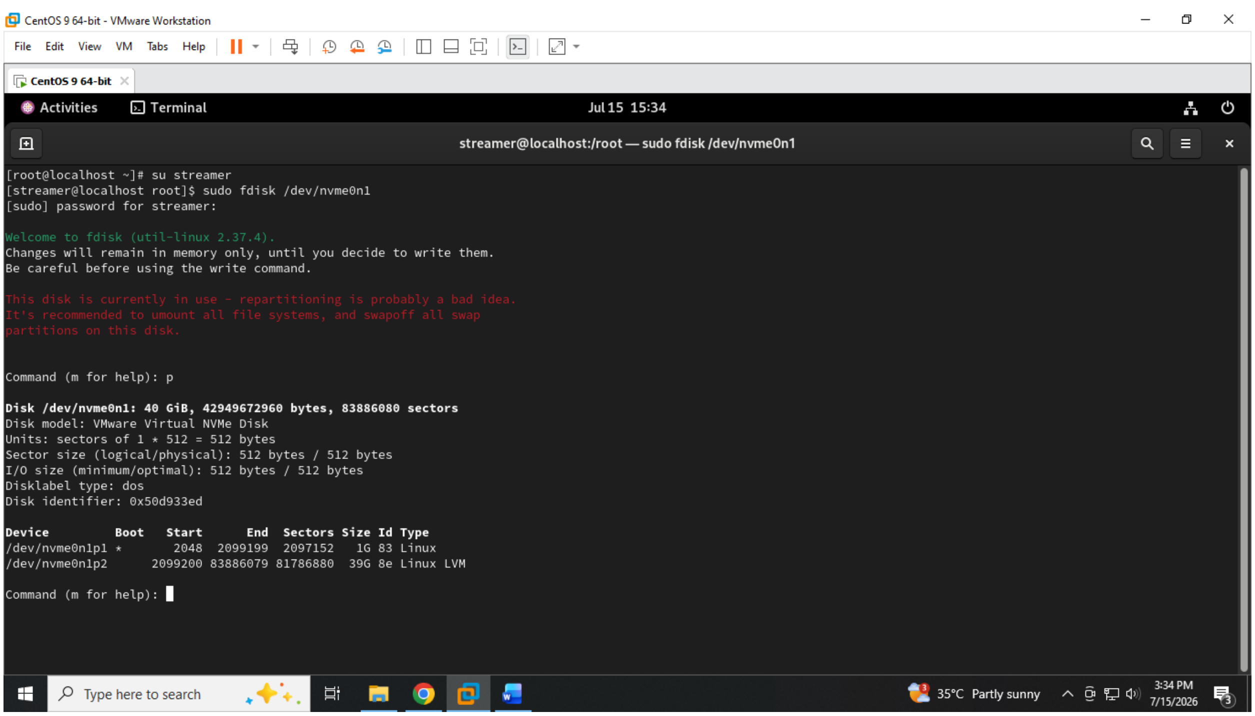
Q.2 Give your current user sudo privileges by editing the /etc/sudoers file using visudo. Test by running a command that requires sudo, such as updating the package list.

ANS :





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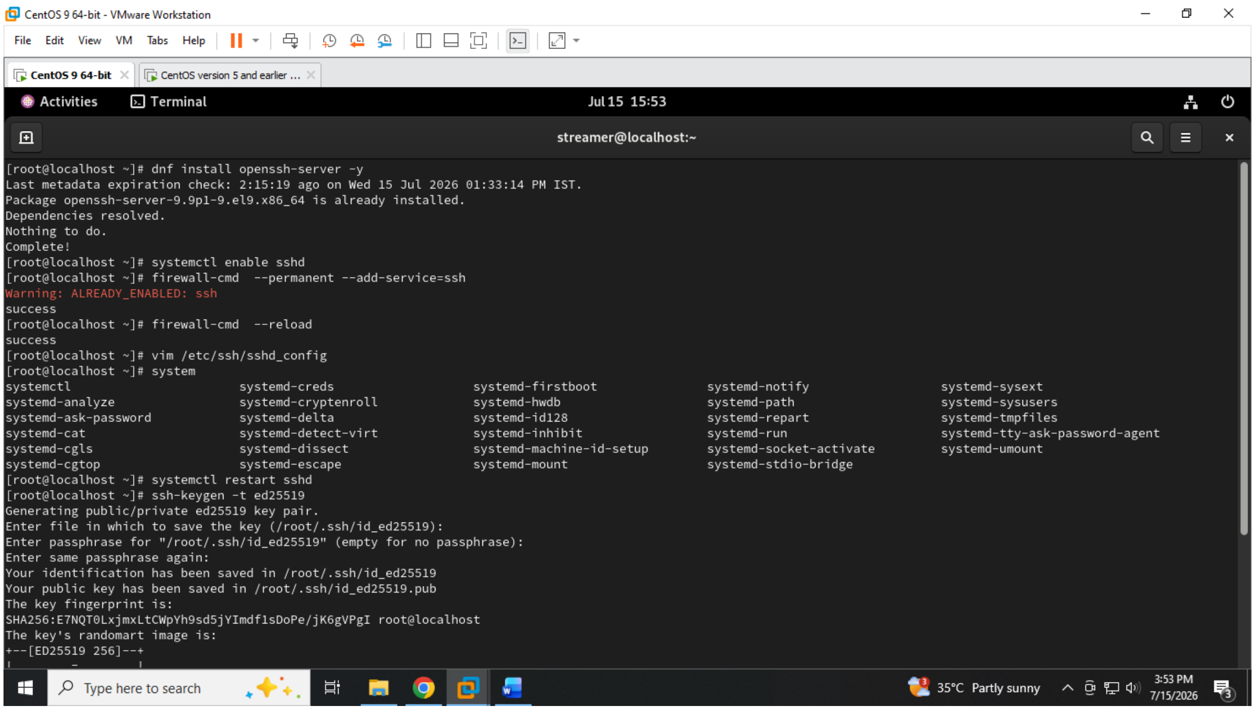
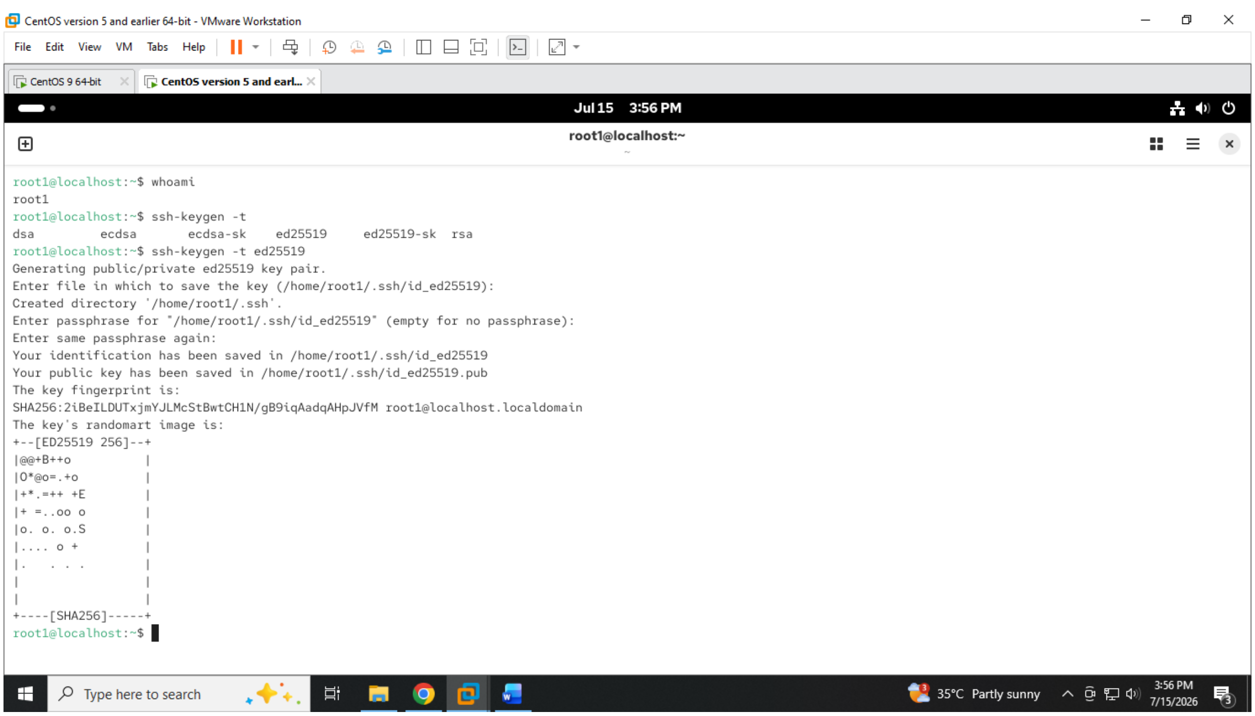
Q.3 Set up SSH key-based authentication: generate an SSH key pair using ssh-keygen, then copy your public key to a remote Linux server (or a local VM) using ssh-copy-id. Disable password authentication for SSH in /etc/ssh/sshd_config and restart the SSH service. Hint: Look for the PasswordAuthentication setting in the SSH config file.

ANS :





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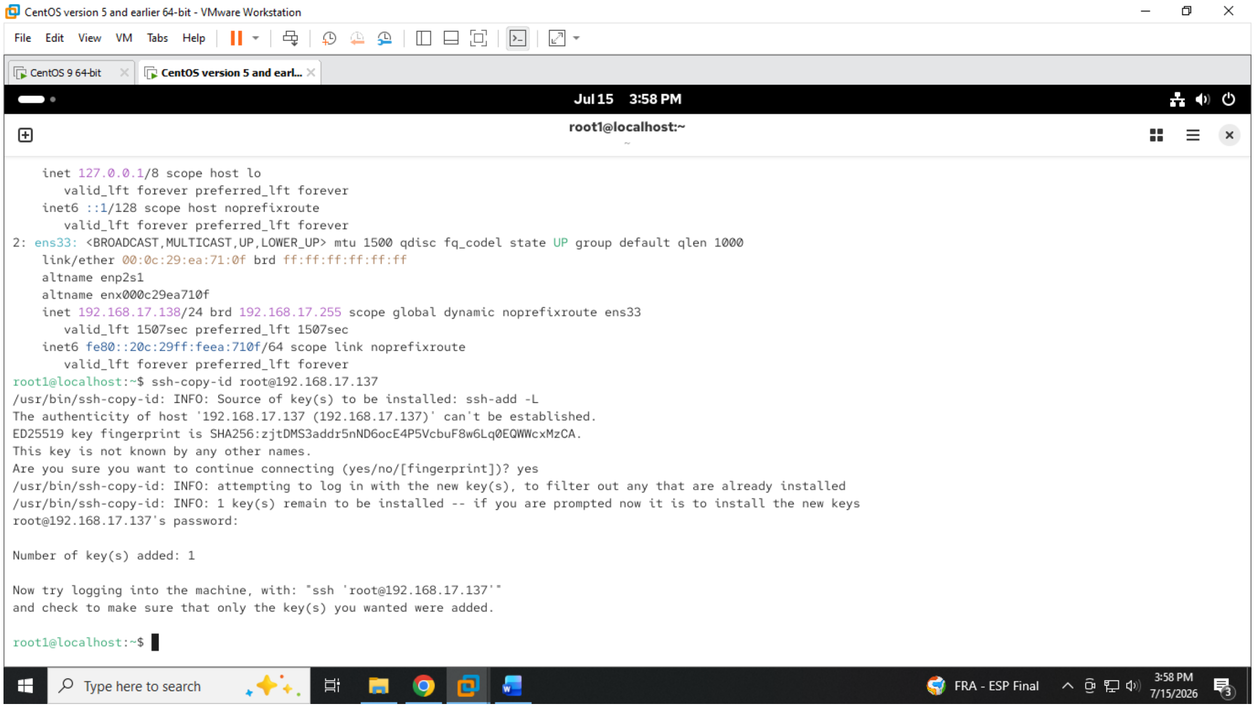

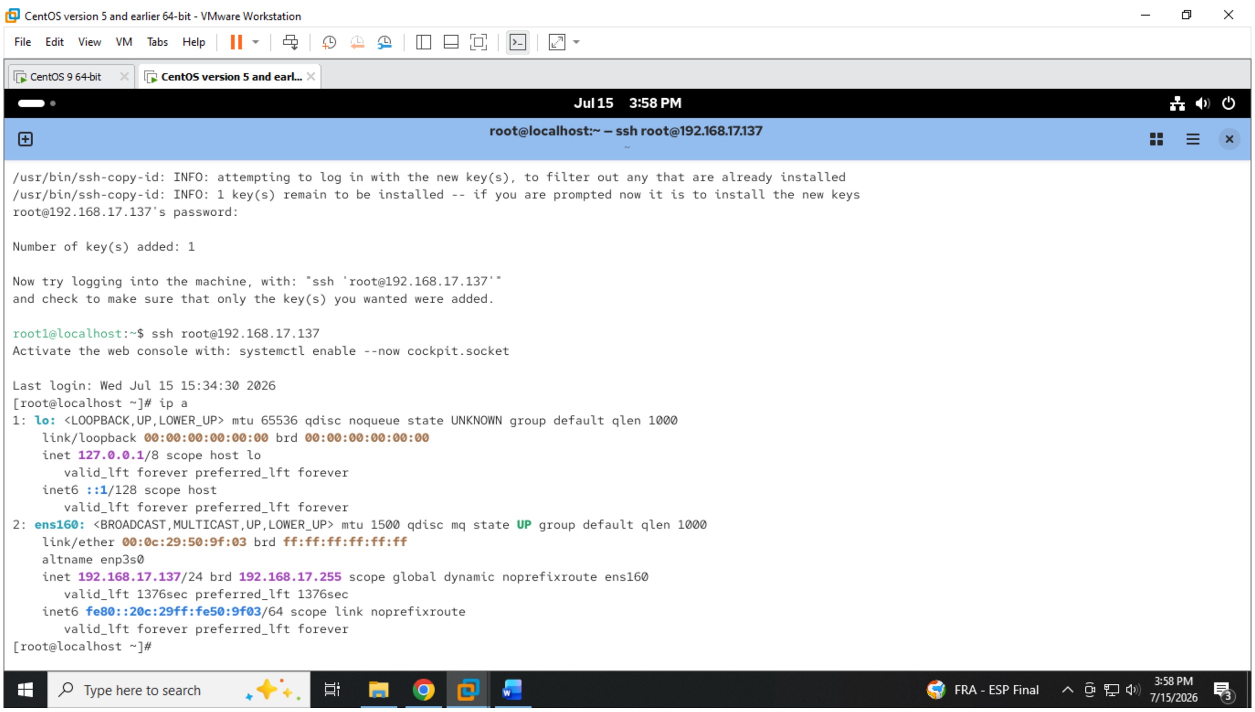

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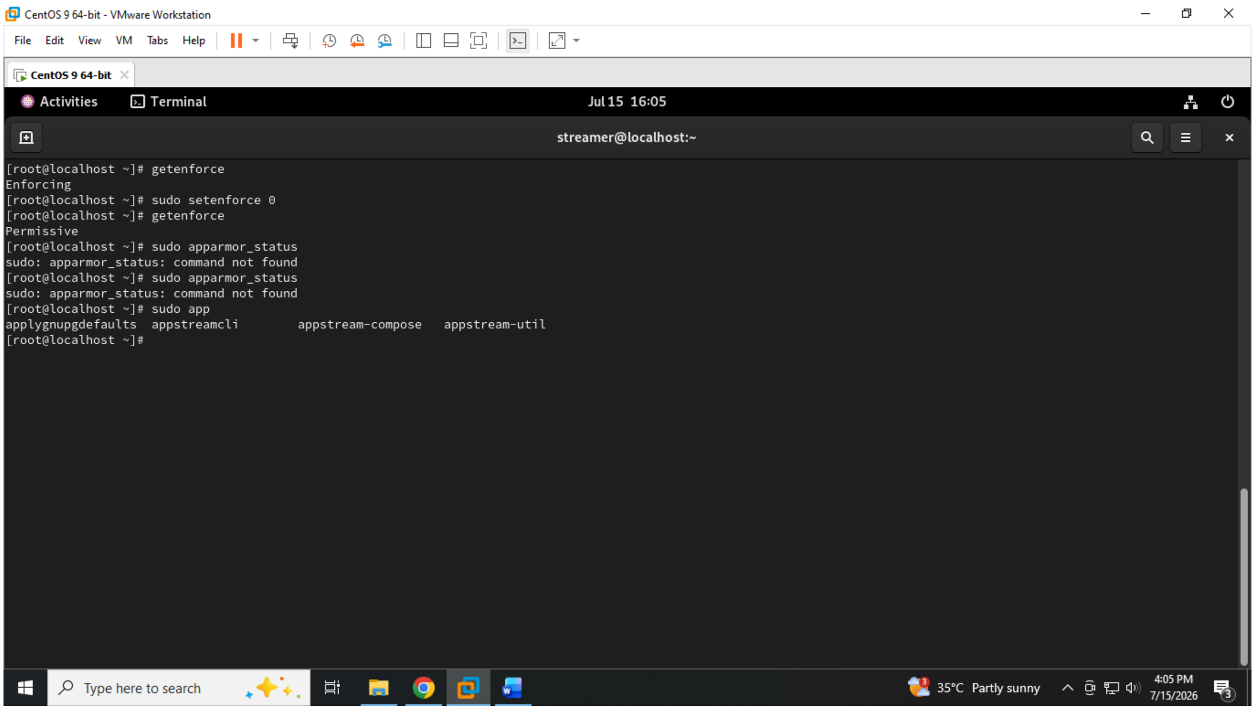
Q.4 Check if SELinux or AppArmor is enabled on your system. If SELinux is active, use the getenforce command to display its current mode, and switch it to permissive mode. If AppArmor is active, list all enforced profiles using sudo apparmor_status.

ANS :



Q.5 Use ChatGPT to explain the difference between SELinux and AppArmor in one paragraph, then write a summary in your own words and mention one real-world app (like Zomato or Paytm) that would benefit from these security modules.

ANS : **Difference between SELinux and AppArmor (explained by ChatGPT)**

SELinux (Security-Enhanced Linux) and AppArmor are both Linux security modules used to enforce access control policies and protect systems from unauthorized actions. SELinux uses a **label-based security model**, where files, processes, and resources are assigned security labels and access decisions are made according to detailed policies. AppArmor uses a **path-based security model**, where security rules are defined based on the file paths that applications can access. SELinux is generally considered more complex and fine-grained, while AppArmor is often easier to configure and manage. Both help reduce the impact of security vulnerabilities by restricting what applications and users are allowed to do.

Summary in my own words

SELinux and AppArmor are security tools that add an extra layer of protection to Linux systems. They control application permissions so that even if an application is compromised, the attacker has limited ability to access sensitive files or perform harmful actions. SELinux provides stronger and more detailed control but requires more knowledge to manage, while AppArmor provides simpler application-based protection that is easier to set up.

A real-world application such as **Zomato** could benefit from these security modules because its servers handle sensitive information such as user accounts, payment details, and order data. SELinux or AppArmor could help restrict database services, payment systems, and web applications so that a security issue in one component does not compromise the entire system.